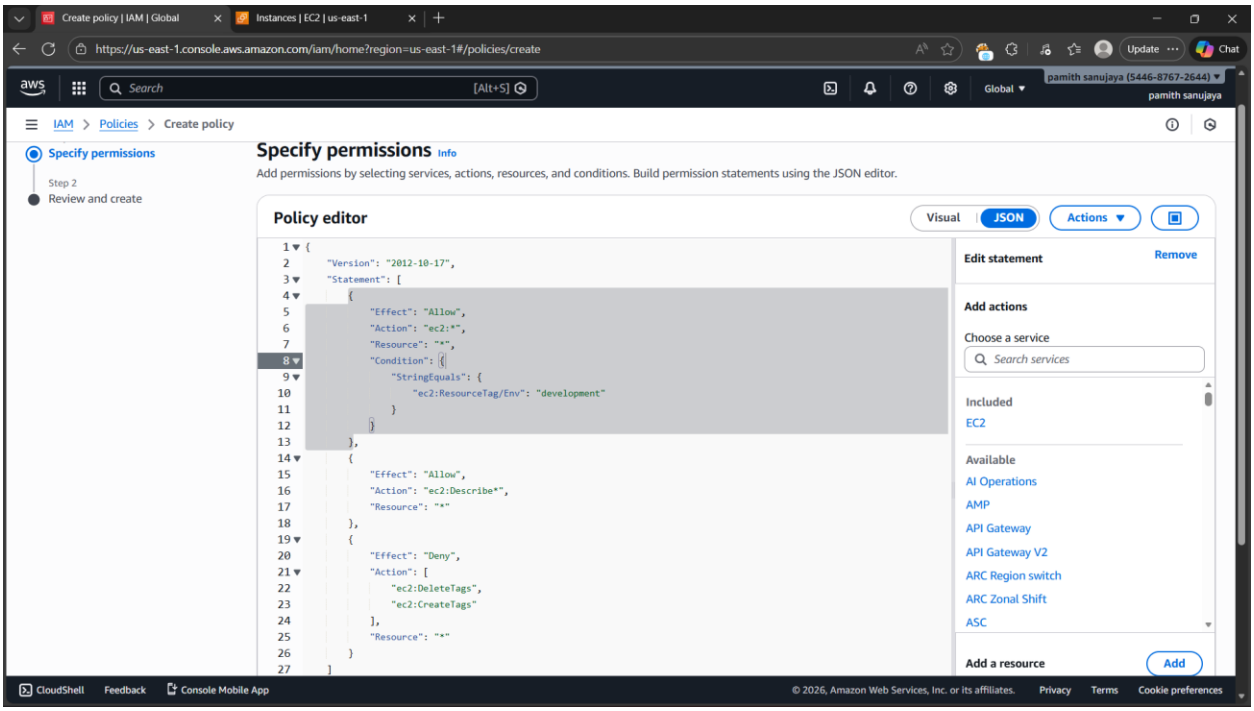
AWS cloud security with IAM

This project provides a hands on demonstration of AWS Identity and Access Management (IAM) security best practices. Through a step by step walkthrough, I will:

- **Create and configure an IAM user and user group,**
- **Assign granular permissions to the IAM user, and**
- **Illustrate the security boundaries by demonstrating what happens when an IAM user attempts to perform actions outside their assigned permissions.**

- 
Create policy | IAM | Global
Instances | EC2 | us-east-1
https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/policies/create
IAM > Policies > Create policy
Specify permissions
Step 2
Review and create
Specify permissions Info
Add permissions by selecting services, actions, resources, and conditions. Build permission statements using the JSON editor.
Policy editor
Visual
JSON
Actions
{
"Version": "2012-10-17",
"Statement": [
{
"Effect": "Allow",
"Action": "ec2:*",
"Resource": "*",
"Condition": {
"StringEquals": {
"ec2:ResourceTag/Env": "development"
}
}
},
{
"Effect": "Allow",
"Action": "ec2:Describe*",
"Resource": "*"
},
{
"Effect": "Deny",
"Action": [
"ec2:DeleteTags",
"ec2:CreateTags"
],
"Resource": "*"
}
]
Edit statement
Remove
Add actions
Choose a service
Search services
Included
EC2
Available
AI Operations
AMP
API Gateway
API Gateway V2
ARC Region switch
ARC Zonal Shift
ASC
Add a resource
Add
CloudShell
Feedback
Console Mobile App
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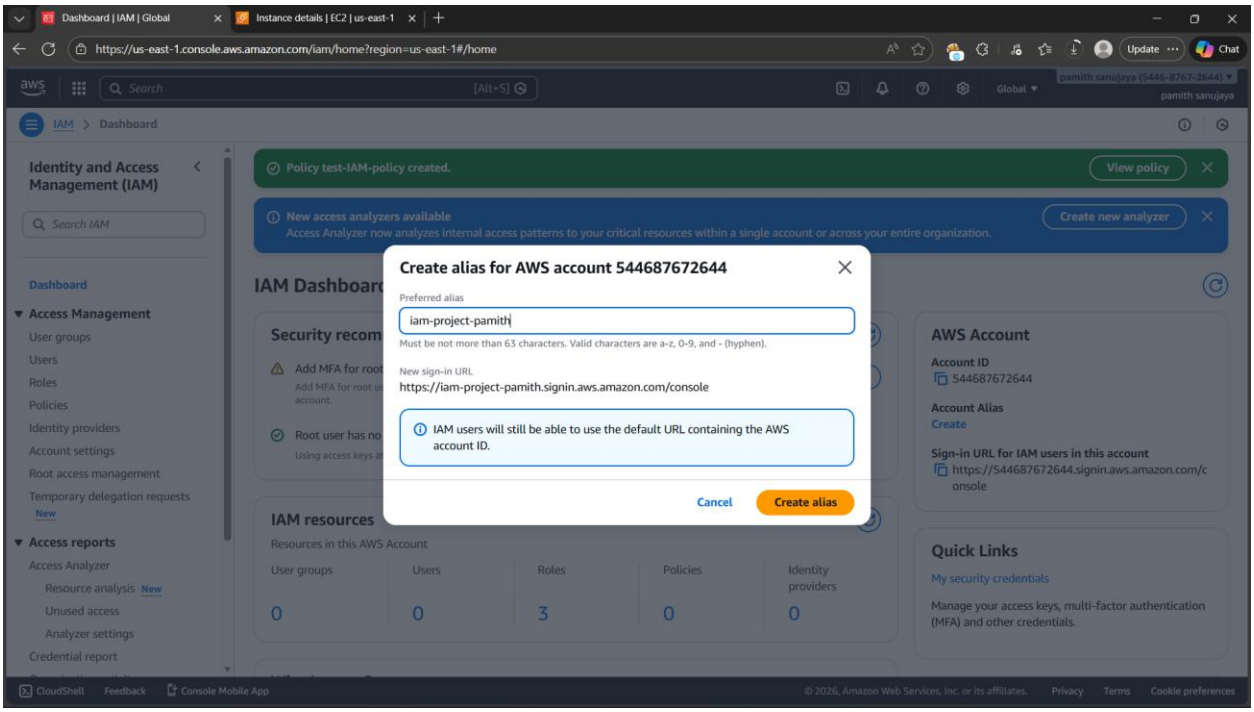
Dashboard | IAM | Global
Instance details | EC2 | us-east-1
https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/home
IAM > Dashboard
Identity and Access Management (IAM)
Search IAM
Dashboard
Access Management
User groups
Users
Roles
Policies
Identity providers
Account settings
Root access management
Temporary delegation requests
New
Access reports
Access Analyzer
Resource analysis New
Unused access
Analyzer settings
Credential report
Policy test-IAM-policy created.
View policy
New access analyzers available
Access Analyzer now analyzes internal access patterns to your critical resources within a single account or across your entire organization.
Create new analyzer
IAM Dashboard
Security recom
Add MFA for root
Root user has no
IAM resources
Resources in this AWS Account
User groups
Users
Roles
Policies
Identity providers
0
0
3
0
0
Create alias for AWS account 544687672644
Preferred alias
iam-project-pamith
Must be not more than 63 characters. Valid characters are a-z, 0-9, and - (hyphen).
New sign-in URL
https://iam-project-pamith.signin.aws.amazon.com/console
IAM users will still be able to use the default URL containing the AWS account ID.
Cancel
Create alias
AWS Account
Account ID
544687672644
Account Alias
Create
Sign-in URL for IAM users in this account
https://544687672644.signin.aws.amazon.com/console
Quick Links
My security credentials
Manage your access keys, multi-factor authentication (MFA) and other credentials.
CloudShell
Feedback
Console Mobile App
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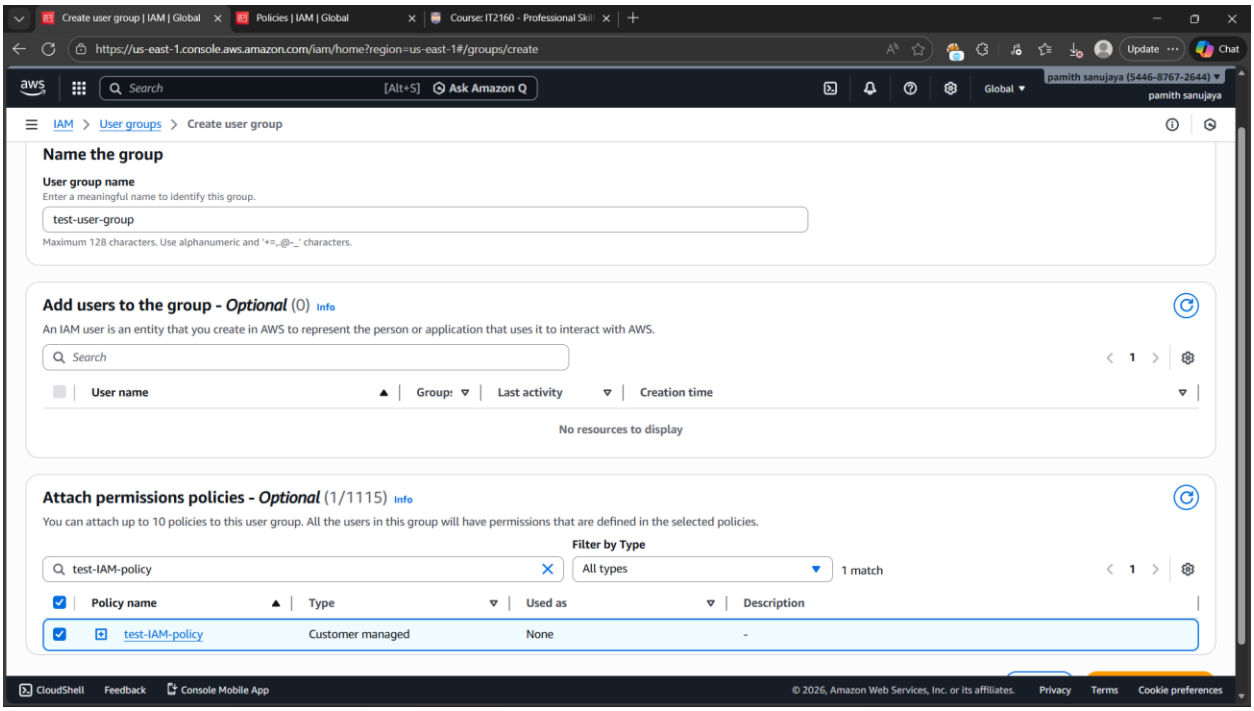
Create user group | IAM | Global
Policies | IAM | Global
Course: IT2160 - Professional Skil
https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups/create
Update
Chat
Search
[Alt+S]
Ask Amazon Q
Global
pamith sanujaya (5446-8767-2644)
pamith sanujaya
IAM > User groups > Create user group
Name the group
User group name
Enter a meaningful name to identify this group.
test-user-group
Maximum 128 characters. Use alphanumeric and '+=,.@-_' characters.
Add users to the group - Optional (0) Info
An IAM user is an entity that you create in AWS to represent the person or application that uses it to interact with AWS.
Search
User name
Groups
Last activity
Creation time
No resources to display
Attach permissions policies - Optional (1/1115) Info
You can attach up to 10 policies to this user group. All the users in this group will have permissions that are defined in the selected policies.
Filter by Type
test-IAM-policy
All types
1 match
Policy name
Type
Used as
Description
test-IAM-policy
Customer managed
None
-
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Feedback
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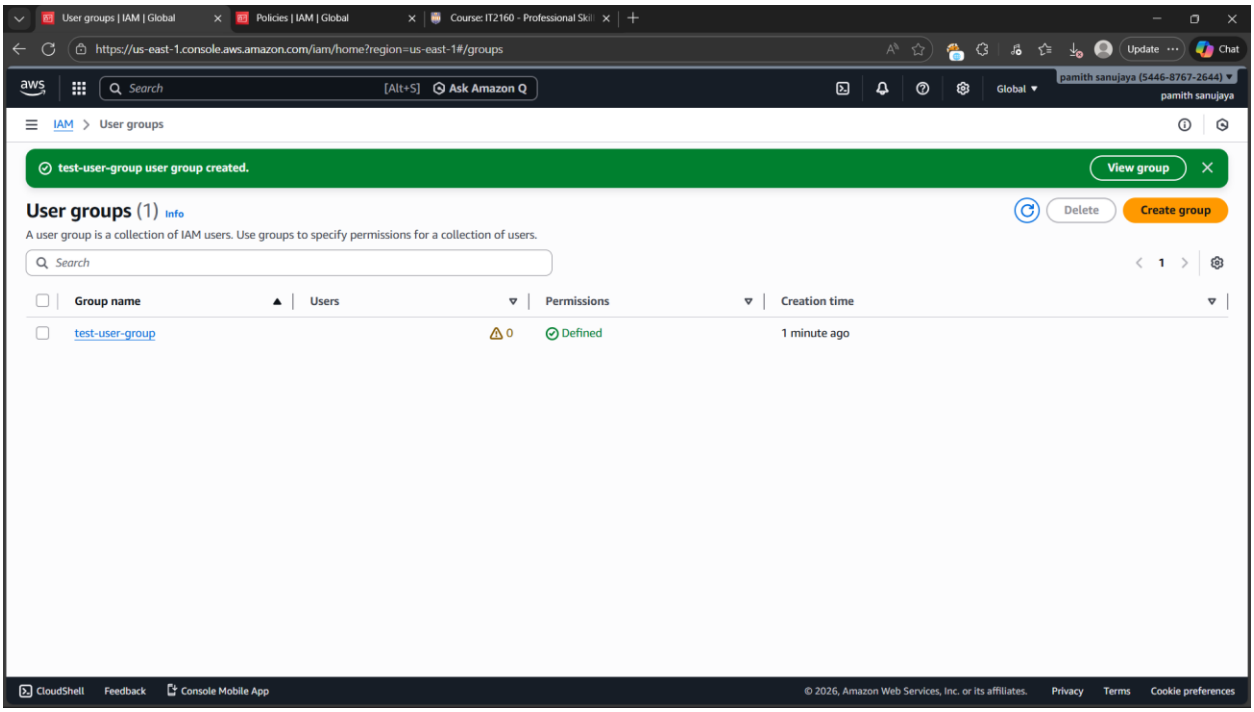
User groups | IAM | Global
Policies | IAM | Global
Course: IT2160 - Professional Skil
https://us-east-1.console.aws.amazon.com/iam/home?region=us-east-1#/groups
Update
Chat
Search
[Alt+S]
Ask Amazon Q
Global
pamith sanujaya (5446-8767-2644)
pamith sanujaya
IAM > User groups
test-user-group user group created.
View group
User groups (1) Info
Delete
Create group
A user group is a collection of IAM users. Use groups to specify permissions for a collection of users.
Search
Group name
Users
Permissions
Creation time
test-user-group
0
Defined
1 minute ago
CloudShell
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aws | Search [Alt+S] | Global | pamith sanujaya (5446-8767-2644) | pamith%20sanujaya

IAM > Users

Identity and Access Management (IAM)

Search IAM

- Dashboard
- **Access Management**
  - User groups
  - Users
  - Roles
  - Policies
  - Identity providers
  - Account settings
  - Root access management
  - Temporary delegation requests New

Users (0) Info

An IAM user is an identity with long-term credentials that is used to interact with AWS in an account.

Delete | Create user

Search

User name	Path	Group:	Last activity	MFA	Password age	Console last sign-in	Ac

No resources to display

IAM > Users > Create user

Step 1 Specify user details
Step 2 Set permissions
Step 3 Review and create
Step 4 Retrieve password

Specify user details

User details

User name

test-user

The user name can have up to 64 characters. Valid characters: A-Z, a-z, 0-9, and + = , . @ _ - (hyphen)

☑ Provide user access to the AWS Management Console - *optional*
In addition to console access, users with SignInLocalDevelopmentAccess permissions can use the same console credentials for programmatic access without the need for access keys.

Console password

- (●) Autogenerated password
  You can view the password after you create the user.
- ( ) Custom password
  Enter a custom password for the user.

- Must be at least 8 characters long
- Must include at least three of the following mix of character types: uppercase letters (A-Z), lowercase letters (a-z), numbers (0-9), and symbols ! @ # $ % ^ & * ( ) _ + - (hyphen) = [ ] { } | '

☐ Show password

☐ Users must create a new password at next sign-in - Recommended
Users automatically get the IAMUserChangePassword policy to allow them to change their own password.

ⓘ If you are creating programmatic access through access keys or service-specific credentials for AWS CodeCommit or Amazon Keyspaces, you can generate them after you create this IAM user. Learn more

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IAM > Users > Create user

Step 1 Specify user details
Step 2 Set permissions
Step 3 Review and create
Step 4 Retrieve password

Set permissions

Add user to an existing group or create a new one. Using groups is a best-practice way to manage user's permissions by job functions. Learn more

Permissions options

- (●) **Add user to group**
  Add user to an existing group, or create a new group. We recommend using groups to manage user permissions by job function.
- ( ) **Copy permissions**
  Copy all group memberships, attached managed policies, and inline policies from an existing user.
- ( ) **Attach policies directly**
  Attach a managed policy directly to a user. As a best practice, we recommend attaching policies to a group instead. Then, add the user to the appropriate group.

User groups (1/1)

Create group

Search

Group name	Users	Attached policies	Created
test-user-group	0	test-IAM-policy	2026-02-12 (3 days ago)

▸ **Set permissions boundary - *optional***

Cancel | Previous | Next

IAM > Users > Create user

User created successfully

You can view and download the user's password and email instructions for signing in to the AWS Management Console.

View user

Step 1
Specify user details

Step 2
Set permissions

Step 3
Review and create

Step 4
Retrieve password

Retrieve password

You can view and download the user's password below or email users instructions for signing in to the AWS Management Console. This is the only time you can view and download this password.

Console sign-in details

Email sign-in instructions

Console sign-in URL
https://iam-project-pamith.signin.aws.amazon.com/console

User name
test-user

Console password
********** Show

Cancel | Download .csv file | Return to users list

Amazon Web Services Sign-In

https://eu-north-1.signin.aws.amazon.com/oauth?client_id=arn%3Aaws%3Asignin%3A%3A%3Aconsole%2Fcanvas&code_challenge=LG3xxzlx2a2JzChdGS8-AuiyusCaguTaoTxQs20qlI...

Provide feedback | Multi-session disabled | English

aws

IAM user sign in

Account ID or alias (Don't have?)
iam-project-pamith

Remember this account

IAM username
test-user

Password
••••••••

Show Password | Having trouble?

Sign in

Sign in using root user email

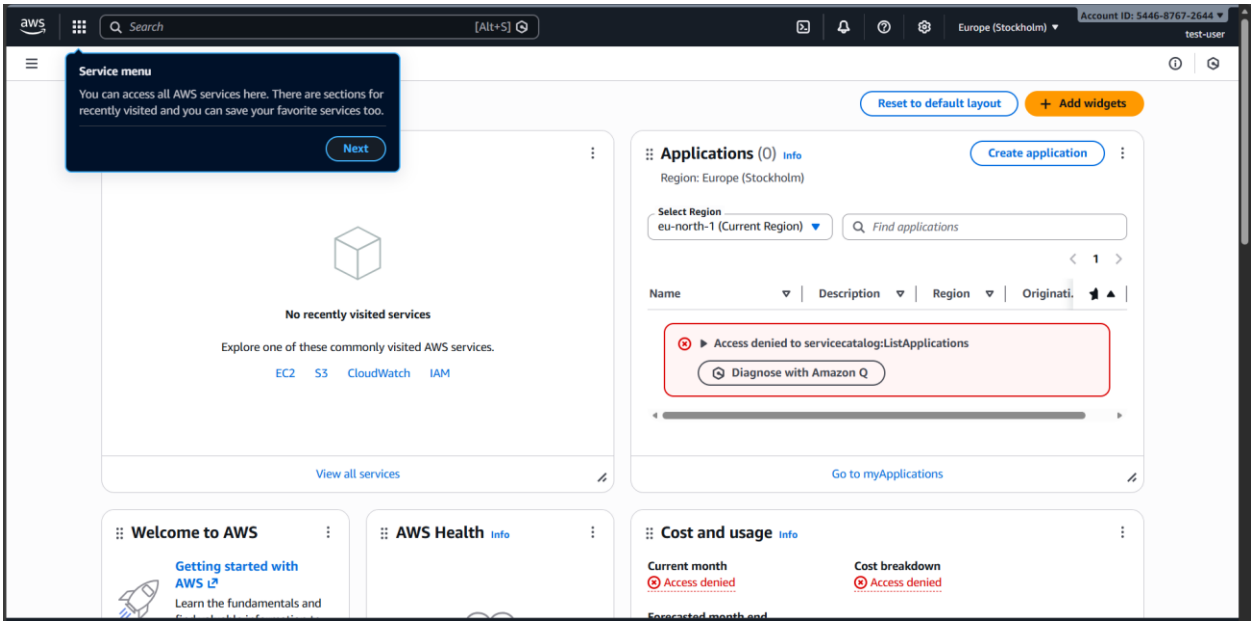
Create a new AWS account

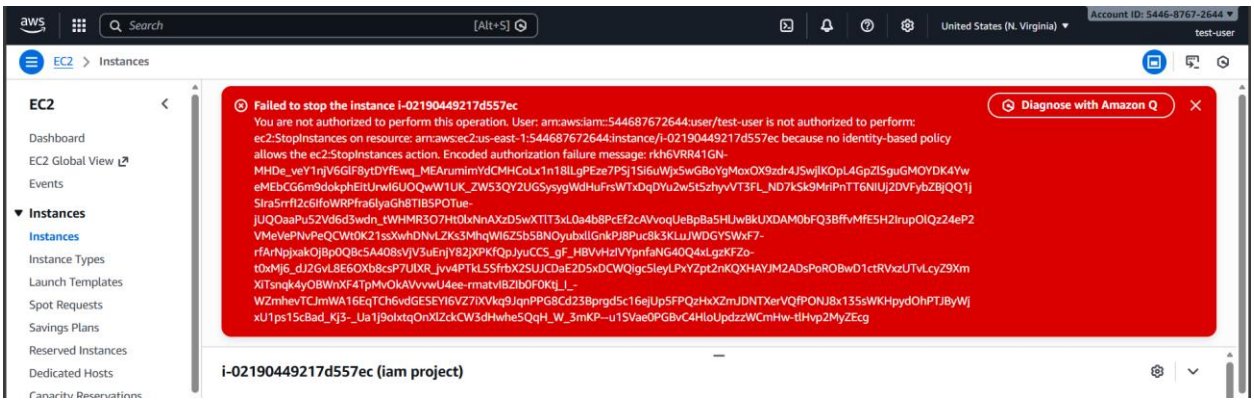
By continuing, you agree to AWS Customer Agreement or other agreement for AWS services, and the Privacy Notice. This site uses essential cookies. See our Cookie Notice for more information.

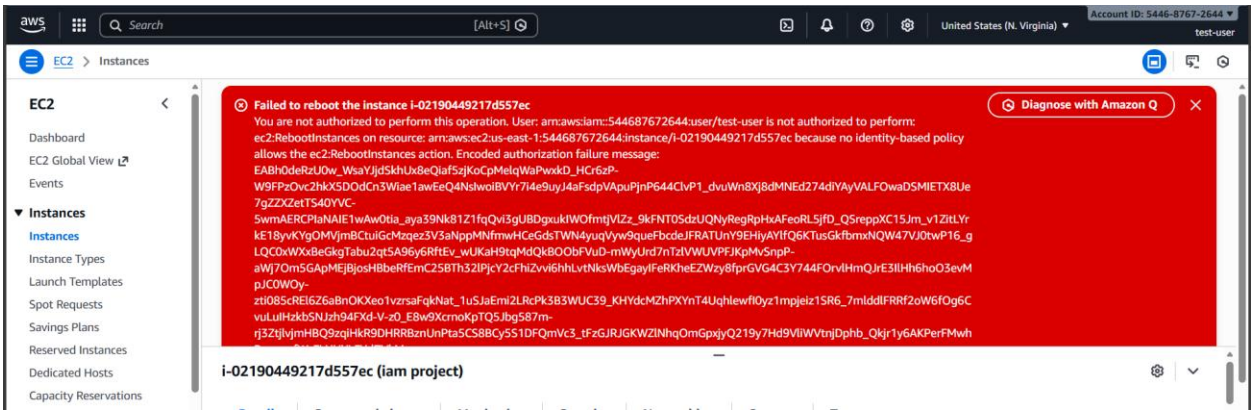
Amazon Lightsail

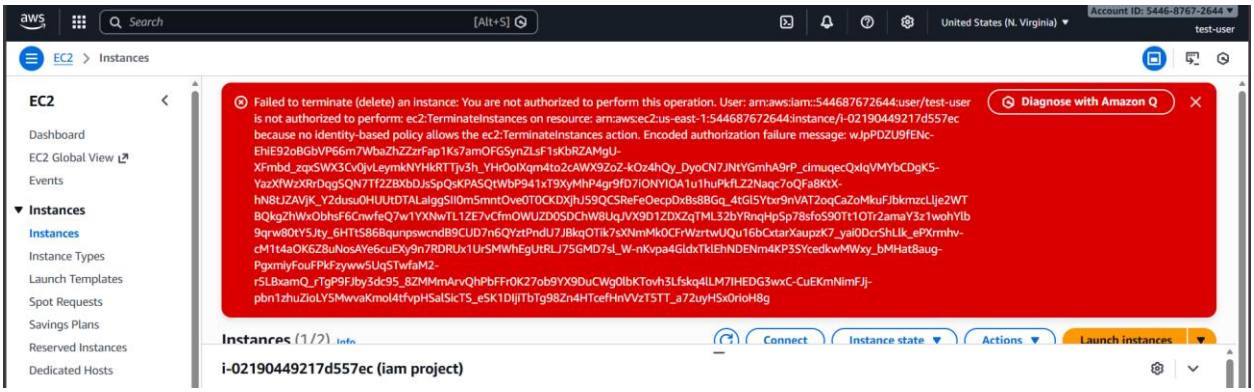
Lightsail is the easiest way to get started on AWS

Learn more »


Search
[Alt+S]
Europe (Stockholm)
Account ID: 5446-8767-2644
test-user
Service menu
You can access all AWS services here. There are sections for recently visited and you can save your favorite services too.
Next
Reset to default layout
+ Add widgets
Applications (0) Info
Create application
Region: Europe (Stockholm)
Select Region
eu-north-1 (Current Region)
Find applications
Name
Description
Region
Originati.
Access denied to servicecatalog:ListApplications
Diagnose with Amazon Q
No recently visited services
Explore one of these commonly visited AWS services.
EC2
S3
CloudWatch
IAM
View all services
Go to myApplications
Welcome to AWS
Getting started with AWS
Learn the fundamentals and
AWS Health Info
Cost and usage Info
Current month
Access denied
Cost breakdown
Access denied


Search
[Alt+S]
United States (N. Virginia)
Account ID: 5446-8767-2644
test-user
EC2 > Instances
EC2
Dashboard
EC2 Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Failed to stop the instance i-02190449217d557ec
You are not authorized to perform this operation. User: arn:aws:iam::544687672644:user/test-user is not authorized to perform: ec2:StopInstances on resource: arn:aws:ec2:us-east-1:544687672644:instance/i-02190449217d557ec because no identity-based policy allows the ec2:StopInstances action. Encoded authorization failure message: rkh6VRR41GN-MHDe_veY1njV6GlF8ytDYfEwq_MEArumimYdCMHCoLx1n18lLgPEze7PSj1Si6uWjx5wGBoYgMoxOX9zdr4JSwjlKOpL4GpZlSguGMOYDK4YweMEbCG6m9dokphEitUrwI6UOQwW1UK_ZW53QY2UGSysygWdHuFrsWTxDqDYu2w5t5zhyvVT3FL_ND7kSk9MriPnTT6NlUj2DVFybZBjQQ1jSlra5rrfI2c6IfoWRPfra6lyaGh8TIB5POTue-jUQOaaPu52Vd6d3wdn_tWHMR3O7Ht0lxNnAXzD5wXTlT3xL0a4b8PcEf2cAVvoqUeBpBa5HLJwBkUXDAM0bFQ3BffvMfE5H2IrupOlQz24eP2VMeVePNvPeQCWt0K21ssXwhDNvLZKs3MhqWI6Z5b5BNOyubxllGnkPJ8Puc8k3KLuJWDGYSWxF7-rfArNpjxakOjBp0QBc5A408sVjV3uEnjY82jXPKfQpJyuCCS_gF_HBVvHzIVYpnfaNG40Q4xLgzKFZo-t0xMj6_dJ2GvL8E6OXb8csP7UlXR_jvv4PTkL5SfrbX2SUJCDaE2D5xDCWQigc5leyLPxYZpt2nKQXHAYJM2ADsPoROBwD1ctRVxzUTvLcyZ9XmXiTsnqk4yOBWnXF4TpMvOkAVvwU4ee-rmatvIBZIb0F0Ktj_I_-WZmhevTCJmWA16EqTCh6vdGESEYI6VZ7iXVkq9JqnPPG8Cd23Bprgd5c16ejUp5FPQzHxXZmJDNTXerVQfPONJ8x135sWKHpydOhPTJByWjxU1ps15cBad_Kj3-_Ua1j9oIxtqOnXlZckCW3dHwhe5QqH_W_3mKP--u1SVae0PGBvC4HloUpdzzWCmHw-tlHvp2MyZEcg
Diagnose with Amazon Q
i-02190449217d557ec (iam project)





This practical approach highlights the importance of least privilege access and effective permission management in AWS environments.